Evaluating Generative Models

CS236: Deep Generative Models — Lecture 15

Seungho Go

2026-04-05

Outline

1. **Background**: model families & why evaluation is hard
2. **Density estimation & compression**
   - Likelihood, perplexity, limits of compression
   - ELBO for VAEs
   - KDE and importance sampling
3. **Sample quality**: HYPE, IS, FID, KID, HEIM
4. **Latent representations**: clustering, compression, disentanglement
5. **Language models**: prompting, fine-tuning, HELM, BigBench

Mid-quarter Recap: Generative Model Families

Density models

- AR: $p_\theta(\mathbf{x}) = \prod_i p_\theta(x_i | x_{<i})$
- Flow: $p_\theta(\mathbf{x}) = p(\mathbf{z}) |\det J_{f_\theta}|$
- VAE: $p_\theta(\mathbf{x}) = \int p(\mathbf{z}) p_\theta(\mathbf{x}|\mathbf{z})\, d\mathbf{z}$
- EBM: $p_\theta(\mathbf{x}) = e^{f_\theta(\mathbf{x})} / Z(\theta)$

Implicit / score models

- GAN: $\mathbf{x} = g_\theta(\mathbf{z}),\ \mathbf{z} \sim p(\mathbf{z})$
- Score-based: learn $\nabla_{\mathbf{x}} \log p(\mathbf{x})$

Training objectives

- KL div. (MLE): AR, Flow
- ELBO: VAE
- Contrastive Div.: EBM
- f-div., Wasserstein: GAN
- Fisher div. (score matching): EBM, score-based
- NCE: EBM

Why Is Evaluation So Hard?

- **Discriminative models**: evaluate with a task-specific loss on held-out test data
  $\rightarrow$ straightforward (e.g., top-1 accuracy)
- **Generative models:** no single agreed-upon task
- The right metric depends entirely on the **end goal**:

Goal	Metric family
Density estimation	Log-likelihood, perplexity
Compression	Bits-per-dimension
Sample quality	IS, FID, KID, human eval
Latent representations	Clustering, PSNR, disentanglement

- A model that excels at sampling can fail at density estimation, and vice versa

Evaluation: Likelihood & Compression

Maximum likelihood $\Leftrightarrow$ minimizing average code length (Shannon, 1948):

$$\text{avg. code length} = -E_{x\sim p_{\text{data}}}\left[\log_2 p_\theta(x)\right]$$

- Morse code intuition: frequent symbols (E, A, $\cdots$) get *short* codes; rare symbols get *long* codes
- Optimal code assigns length $-\log_2 p(x)$ to each x
- **Perplexity**: $\mathrm{PPL} = 2^{-E[\log_2 p_\theta(x)]}$ — standard LM metric
- Hutter Prize: compress 1 GB of Wikipedia $\Rightarrow$ compression ability $\approx$ intelligence
- LLMs already beat humans: $\approx$0.94 bits/char vs. $\approx$1.3 bits/char

Practical: arithmetic coding achieves the Shannon limit; perplexity is its scaled version.

Limitation: Not All Bits Are Equal

- Compression treats every bit identically
- But in practice, **not all information has the same value**:
  - An image pixel changing from (128,128,128) to (129,128,128): costs bits, but we might not care
  - The label "cat" vs. "dog" also costs bits, but we care a lot
- **Bigger problem**: many generative models have intractable likelihoods
  - VAE: $\log p_\theta(\mathbf{x}) = \log \int p(\mathbf{z}) p_\theta(\mathbf{x}|\mathbf{z})\, d\mathbf{z}$ — intractable integral
  - GAN: no density defined at all
  - EBM: $Z(\theta)$ is unknown
- For VAEs, we can compare **ELBO** (a lower bound on log-likelihood)
  For GANs, we need entirely different approaches

ELBO: Evidence Lower Bound for VAEs

VAE likelihood is **intractable** (integral over all $\mathbf{z}$):

$$\log p_\theta(\mathbf{x}) = \log \int p(\mathbf{z})\, p_\theta(\mathbf{x}|\mathbf{z})\, d\mathbf{z}$$

Introduce approximate posterior $q_\phi(\mathbf{z}|\mathbf{x})$ and apply **Jensen's inequality**:

$$\log p_\theta(\mathbf{x}) \;\geq\; \underbrace{E_{q_\phi(\mathbf{z}|\mathbf{x})}\big[\log p_\theta(\mathbf{x}|\mathbf{z})\big] \;-\; D_{\mathrm{KL}}\big(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, p(\mathbf{z})\big)}_{\mathrm{ELBO}(\theta,\phi;\,\mathbf{x})}$$

- **Reconstruction term**: how well can we decode $\mathbf{x}$ from $\mathbf{z}$?
- **Regularization term**: how close is q_ϕ to the prior $p(\mathbf{z})$?
- Gap $= D_{\mathrm{KL}}(q_\phi \| p_\theta(\mathbf{z}|\mathbf{x})) \geq 0 \;\Rightarrow$ ELBO is always a *lower bound*
- Maximising ELBO simultaneously raises likelihood and improves q_ϕ

Estimating Density from Samples: KDE & AIS

When only samples are available (e.g., GAN), use **KDE**:

$$\hat{p}(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n} K\left(\frac{\mathbf{x}-\mathbf{x}^{(i)}}{\sigma}\right), \quad K(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2}$$

Bandwidth σ controls smoothness; tuned via cross-validation. Limitation: curse of dimensionality makes KDE unreliable for images.

When the model has a latent structure (VAE, EBM), use **Importance Sampling**:

$$p_\theta(\mathbf{x}) = E_{p(\mathbf{z})}\big[p_\theta(\mathbf{x}|\mathbf{z})\big] \xrightarrow{\text{proposal } q_\phi} \text{IWAE (tighter than ELBO)}$$

Annealed IS (AIS): interpolate intermediate distributions from $p(\mathbf{z})$ to $p(\mathbf{z}|\mathbf{x}) \Rightarrow$ unbiased estimate of Z_2/Z_1.

Fundamental limit: for models with *samples only*, unbiased density estimation is impossible $\Rightarrow$ need sample-quality metrics.

Sample Quality: Human Evaluation (HYPE)

- **Gold standard**: ask humans to distinguish real from generated
- **HYPE** (Human eYe Perceptual Evaluation, Zhou et al. 2019):
  - $\text{HYPE}_{\text{time}}$: minimum viewing time needed for accurate classification (higher = better)
  - HYPE_{∞}: fraction of generated samples that fool humans with unlimited time (higher = better)
- StyleGAN achieves $\sim 50\%$ HYPE_{∞} — half of samples fool even careful humans
- **Limitations**:
  - Expensive, slow, hard to reproduce
  - Cannot penalise *memorisation* of training data
  - Subjective and scale-dependent
- $\Rightarrow$ We need **automatic**, **reproducible** metrics

Inception Score (IS)

Use pretrained classifier $c(y|\mathbf{x})$ (Inception-v3 on ImageNet). Two criteria:

Sharpness S

Each sample is classified confidently

$c(y|\mathbf{x})$ has *low* entropy

$$S = \exp\Big(E_{\mathbf{x}}\Big[\sum_y c \log c\Big]\Big)$$

Diversity D

Many different classes are generated

marginal $c(y) = E_{\mathbf{x}}[c(y|\mathbf{x})]$ has *high* entropy

$$D = \exp\Big(-E_{\mathbf{x}}\Big[\sum_y c(y|\mathbf{x}) \log c(y)\Big]\Big)$$

$$\boxed{IS = D \times S = \exp\Big(E_{\mathbf{x}}\left[D_{\mathrm{KL}}(c(y|\mathbf{x}) \,\|\, c(y))\right]\Big)}$$

Higher IS = better. Limitation: never looks at real data — cannot detect distribution shift.

Fréchet Inception Distance (FID)

IS limitation: compares generated samples against themselves, not against real data.

FID fix: extract Inception features, then compare *feature distributions* of real and generated data.

1. Extract feature vectors $\phi(\mathbf{x})$ from Inception-v3 (penultimate layer)
2. Fit Gaussians: $\mathcal{N}(\mu_G, \Sigma_G)$ for generated; $\mathcal{N}(\mu_T, \Sigma_T)$ for real
3. Compute Wasserstein-2 distance between the two Gaussians:

$$\mathrm{FID} = \|\mu_T - \mu_G\|^2 + \mathrm{Tr}\Big(\Sigma_T + \Sigma_G - 2(\Sigma_T \Sigma_G)^{1/2}\Big)$$

Lower FID = better. De facto standard for image generation.

Limitations: Gaussian assumption may not hold; estimator is *biased* (needs many samples).

Kernel Inception Distance (KID)

FID limitation: biased estimator (Gaussian assumption + finite-sample bias).

KID fix: compute **MMD** (Maximum Mean Discrepancy) in Inception feature space — no distributional assumption needed.

$$\mathrm{MMD}(p, q) = E_p[K(\mathbf{x}, \mathbf{x}')] + E_q[K(\mathbf{x}, \mathbf{x}')] - 2\, E_{p,q}[K(\mathbf{x}, \mathbf{x}')]$$

K is a kernel measuring similarity between feature vectors.

	FID	**KID**
Distributional assumption	Gaussian	None
Estimator	Biased	**Unbiased**
Computation	$O(n)$	$O(n^2)$

In practice, FID is more widely used due to lower cost.

Beyond FID: Evaluating Text-to-Image Models (HEIM)

Text-to-image models require evaluating *many* dimensions beyond sample fidelity:

- **Alignment**: does the image match the caption? (CLIP score)
- **Quality**: FID, IS, KID
- **Aesthetics**: visual appeal
- **Originality**: no training data memorisation
- **Fairness**: gender / racial bias
- **Toxicity**: harmful content generation
- **Robustness**: stability under prompt variation
- **Efficiency**: compute cost

HEIM (Stanford, 2023): 26 models $\times$ 29 scenarios $\times$ 33 metrics (automated + human)

`https://crfm.stanford.edu/heim/latest/`

Evaluating Latent Representations

Beyond sampling — generative models also learn **representations of data**.

What makes a "good" latent space? Three perspectives:

1. **Clustering**
   Semantically similar points should be close in latent space
   $\rightarrow$ useful for semi-supervised classification
2. **Compression (Reconstruction)**
   We should be able to reconstruct the input accurately from the latent code
   $\rightarrow$ useful for denoising, super-resolution
3. **Disentanglement**
   Each latent dimension should control one *independent, interpretable* attribute
   $\rightarrow$ user control over generated content

With a downstream task: evaluate directly by task performance.

Clustering & Reconstruction Quality

Clustering: apply k-means to latent space; compare to ground-truth labels

- **Completeness**: all members of a class land in one cluster $[0, 1]$
- **Homogeneity**: each cluster contains members of only one class $[0, 1]$
- **V-measure**: harmonic mean of the two
- `sklearn.metrics`: `completeness_score`, `homogeneity_score`, `v_measure_score`
- Note: labels only used for evaluation; clustering itself is unsupervised

Reconstruction quality: how faithfully can we recover $\mathbf{x}$ from $\mathbf{z}$?

- **MSE**: $\frac{1}{n}\|\mathbf{x} - \hat{\mathbf{x}}\|^2$
- **PSNR**: $10 \log_{10}\left(\mathrm{MAX}^2/\mathrm{MSE}\right)$ higher $=$ better
- **SSIM**: structure similarity index, correlates with human perception

Disentanglement

Ideal: varying z_i changes *exactly one* interpretable attribute, leaving others fixed.

- $z_1 \uparrow \Rightarrow$ face gets older, nothing else changes
- $z_2 \uparrow \Rightarrow$ skin colour changes, nothing else changes

Quantitative metrics (require labelled evaluation data):

- **Beta-VAE metric** (Higgins et al. 2017): accuracy of a linear classifier that predicts which factor of variation was fixed
- **MIG** (Mutual Information Gap): gap in mutual information between the most and second-most relevant latent dimension
- **DCI**: Disentanglement, Completeness, Informativeness
- SAP score, Modularity — see `disentanglement_lib`

Theoretical impossibility: with only unlabelled data, perfect disentanglement cannot be achieved without additional inductive biases.

Language Models: Prompting & Fine-tuning

LMs are generative models: $p_\theta(\text{next word} \mid \text{context})$.

Two adaptation strategies (beyond plain perplexity):

Prompting

No parameter update — craft input to elicit desired output

- Cheap, no labelled data needed
- Sensitive to prompt phrasing
- Example: append `"Sentiment: __"` to a review for sentiment analysis

Fine-tuning

Update parameters on task data

- Higher task performance
- Requires labelled examples & compute
- **Preference-based FT**: show two outputs, collect human preference, fine-tune to increase preferred output probability

Preference-based FT is used to improve: aesthetic quality, safety, fairness.

Holistic LLM Evaluation: HELM & BigBench

Perplexity alone fails to capture what we care about in LLMs.

HELM (Holistic Evaluation of Language Models, Stanford):

- 81 models, 73 scenarios, 65 metrics
- Covers: accuracy, calibration, robustness, **fairness**, **bias**, **toxicity**, efficiency
- `https://crfm.stanford.edu/helm/latest/`

BigBench (Google):

- 200+ tasks: QA, math, reasoning, code, riddles, $\cdots$
- Community-contributed; tests abilities that current LLMs are predicted to *fail* at
- `https://github.com/google/BIG-bench`

Key insight: **no single metric** captures all dimensions — multi-benchmark evaluation is essential.

Summary

Goal	Primary metric	Limitation
Density estimation	Log-likelihood, PPL	Not all bits are equal
Tractable lower bound	ELBO	Only a lower bound
Density from samples	KDE	Breaks in high dim.
Sample quality	IS, FID, KID, HYPE	Metric gaming possible
Latent clustering	V-measure	Needs labels
Reconstruction	PSNR, SSIM	Pixel-level, ignores semanti
Disentanglement	Beta-VAE, MIG, DCI	Theoretically limited
LLM capabilities	HELM, BigBench	Benchmark saturation

Key takeaway: match the metric to your goal.
Blindly optimising any single metric leads to Goodhart's Law.